

AI Literacy Framework and Strategies for Implementation in Developing Nations

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Abstract

Artificial Intelligence (AI) is a transformative force in the 21st century, offering significant potential to revolutionize various sectors, including education. However, the disparity in AI literacy between developed and developing nations poses significant challenges to achieving equitable global progress. This review paper aims to address these challenges by proposing a comprehensive AI literacy framework tailored for developing nations and outlining strategic implementation plans. The paper begins by defining AI literacy and emphasizing its critical importance for individuals and societies, particularly in the context of developing nations. It provides an in-depth analysis of the current state of AI literacy in these regions, identifying key barriers such as limited infrastructure, insufficient educational resources, and policy gaps. A detailed AI literacy framework is proposed, focusing on essential competencies and skills, curriculum integration, and the pivotal role of educator training. Strategies for effective implementation are explored, including policy recommendations for governments and educational institutions, the role of public-private partnerships, and the utilization of online educational platforms to enhance accessibility and inclusivity. The review highlights successful case studies from various developing nations, extracting best practices and lessons that can be adapted and replicated. Methods for assessing the impact of AI literacy programs are discussed, emphasizing the long-term benefits for both individuals and the broader society. Finally, the paper considers future directions and innovations in AI literacy, suggesting ways to improve and adapt the framework to meet evolving needs continuously. By thoroughly reviewing the existing literature and offering actionable strategies, this paper aims to contribute to the global effort to promote AI literacy and ensure that developing nations are not left behind in the AI-driven future.

CCS Concepts

• **Social and professional topics** → Professional topics; Computing education; Computing literacy.

Keywords

Artificial Intelligence (AI), STEM Education, Equity, AI Frameworks



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1 INTRODUCTION

Artificial Intelligence (AI) has emerged as a transformative force in the 21st century, profoundly impacting various sectors of society. AI, broadly defined as the development of computer systems capable of performing tasks that typically require human intelligence, is reshaping how we live, work, and interact with the world around us as per Russell & Norvig, 2021 [1]. The potential of AI to revolutionize diverse sectors, including education, economics, and healthcare, is immense and far-reaching. In education, AI-powered adaptive learning systems can tailor instruction to individual student needs, potentially bridging achievement gaps and enhancing learning outcomes as per Holmes et al., 2019 [2]. The healthcare sector is witnessing AI applications ranging from early disease detection to drug discovery, promising more efficient and personalized medical care as per Topol, 2019 [3]. Economically, AI's impact is equally significant, with estimates suggesting it could add up to \$15.7 trillion to the global economy by 2030 as per PwC, 2020 [4]. Given AI's pervasive influence, the importance of AI literacy cannot be overstated. AI literacy encompasses the knowledge, skills, and attitudes necessary to understand, critically evaluate, and engage with AI technologies as per Long & Magerko, 2020 [5]. It goes beyond mere technical proficiency, encompassing ethical considerations and societal implications. As AI systems become more integrated into our daily lives, AI literacy is becoming as fundamental as traditional literacy and numeracy skills.

The relevance of AI literacy in the 21st century is particularly pronounced in terms of economic and social development. As industries increasingly adopt AI technologies, individuals with AI literacy are better positioned to participate in and contribute to the evolving job market. Moreover, AI literacy empowers citizens to engage in informed discussions about the ethical and societal implications of AI, ensuring that the development and deployment of these technologies align with human values and societal needs as per Floridi et al., 2018 [6]. However, the global landscape of AI literacy is marked by significant disparities between developed and developing nations. While countries like the United States, China, and those in the European Union are making substantial investments in AI education and research, many developing nations lag behind due to various structural and systemic challenges as per Vinuesa et al., 2020 [7]. This disparity raises concerns about the potential for a widening global digital divide, where developing

nations risk falling further behind in the AI-driven global economy. The implications of this AI literacy gap are far-reaching. As AI technologies increasingly drive economic growth and innovation, nations with lower levels of AI literacy may find themselves at a competitive disadvantage in the global marketplace as per Makridakis, 2017 [8]. This could exacerbate existing economic inequalities and limit opportunities for social mobility in developing nations. Furthermore, the lack of AI literacy could hinder these countries' ability to harness AI to address pressing local challenges, such as improving healthcare delivery or enhancing agricultural productivity. Developing nations face a multitude of challenges in achieving widespread AI literacy. Limited infrastructure and technological resources often restrict access to the tools and platforms necessary for AI education. Many of these countries struggle with unreliable internet connectivity and a shortage of computing devices, making it difficult to implement comprehensive AI education programs as per Nye, 2014 [9]. The lack of access to quality education and training programs further compounds the problem. Many educational institutions in developing nations lack the expertise and resources to offer AI-related courses. There is often a shortage of qualified instructors who can effectively teach AI concepts and applications. Additionally, existing curricula may not adequately address AI literacy, focusing instead on more traditional subjects as per Goksel & Bozkurt, 2019 [10].

Insufficient policy support and funding present another significant hurdle. Many developing nations have yet to prioritize AI education in their national policies, resulting in limited allocation of resources for AI literacy initiatives. The absence of a coherent national strategy for AI development and education can lead to fragmented and ineffective efforts to promote AI literacy as per Dutton, 2018 [11]; "An Overview of National AI Strategies and Policies," 2021 [36]. Cultural and linguistic barriers also play a role in impeding AI literacy in developing nations. Much of the available AI educational content is in English, which can be a significant barrier in non-English speaking countries. Moreover, some cultures may have reservations about adopting new technologies, necessitating tailored approaches that align with local values and contexts as per Schiff, 2021 [12].

Given these unique challenges faced by developing nations, there is a pressing need for a tailored AI literacy framework that addresses their specific circumstances. Such a framework must take into account the resource constraints, cultural contexts, and developmental priorities of these nations while providing a comprehensive roadmap for enhancing AI literacy. This paper aims to address this need by proposing a comprehensive AI literacy framework specifically designed for developing nations and outlining strategic implementation plans. By doing so, we seek to contribute to the global effort to promote AI literacy and ensure that developing nations are not left behind in the AI-driven future. The objectives of this paper are multifold. First, we aim to provide a thorough analysis of the current state of AI literacy in developing nations, identifying key barriers and opportunities. Second, we propose a detailed AI literacy framework that focuses on essential competencies and skills, curriculum integration, and the pivotal role of educator training. Third, we explore strategies for effective implementation, including policy recommendations, the role of public-private partnerships, and the utilization of online educational platforms. The

paper is structured as follows: We begin with an in-depth examination of the current state of AI literacy in developing nations. This is followed by the presentation of our proposed AI literacy framework, tailored to address the unique challenges and opportunities in these regions. We then delve into implementation strategies, explore case studies of successful AI literacy initiatives, discuss impact assessment methods, and consider future directions in AI literacy. By providing a comprehensive review of existing literature and offering actionable strategies, this paper aims to serve as a valuable resource for policymakers, educators, and stakeholders in developing nations seeking to enhance AI literacy. It is our hope that this work will contribute to bridging the global AI literacy gap and empower developing nations to harness the full potential of AI for their economic and social development.

2 LITERATURE REVIEW

This section provides a comprehensive review of existing literature related to AI literacy, exploring its various dimensions, global perspectives, and the current state of AI literacy in developing nations. The review highlights key challenges and opportunities and examines the role of educational frameworks and teacher training in promoting AI literacy.

2.1 Overview of AI Literacy

AI literacy has emerged as a crucial competency in our increasingly digitalized world. As per Long & Magerko, 2020 [5] define AI literacy as "the set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace" (p. 2). This definition encompasses not only technical knowledge but also ethical considerations and practical skills.

The components of AI literacy typically include:

- Understanding of core AI concepts and techniques
- Ability to critically evaluate AI systems and their implications
- Ethical awareness and reasoning about AI's societal impact
- Practical skills in using and interacting with AI tools

AI literacy has become particularly important in education and workforce development. As per Pedró et al., 2019 [13] argue, "AI literacy is becoming as fundamental as reading, writing and arithmetic in preparing students for the future" (p. 17). In the workforce, AI literacy is increasingly seen as a key skill for employability and career advancement across various sectors as per Marr, 2018 [14].

2.2 Global Perspectives on AI Literacy

Research on AI literacy globally reveals significant disparities between developed and developing nations. A study by ss per Gabriel et al., 2022 [15] found that while developed countries are rapidly integrating AI education into their curricula, many developing nations are still struggling with basic digital literacy. Successful AI literacy initiatives in developed nations often share common features. For instance, Finland's Elements of AI course, a free online program aimed at demystifying AI for the general public, has been praised for its accessibility and practical approach (Elementsofai.com, 2021). Similarly, the AI4K12 Initiative in the United States

has been effective in developing guidelines for AI education in primary and secondary schools as per Touretzky et al., 2019 [16].

These successful programs typically incorporate:

- Hands-on, project-based learning
- Interdisciplinary approaches that connect AI to various subjects
- Focus on ethical implications and societal impacts of AI
- Partnerships between educational institutions and industry

2.3 Current State of AI Literacy in Developing Nations

The state of AI literacy in developing nations presents a complex picture. A comprehensive report by UNESCO, 2023 [17] highlights that while there is growing awareness of AI's importance, many developing countries lack the infrastructure and resources to implement widespread AI education.

Common challenges identified in the literature include:

- Limited technological infrastructure: Many schools in developing nations lack reliable internet connectivity and computer resources, hindering the implementation of AI education programs as per Nye, 2014 [9].
- Shortage of qualified educators: There is a significant shortage of teachers trained in AI concepts and pedagogies in developing countries as per Goksel & Bozkurt, 2019 [10].
- Curriculum gaps: Many education systems in developing nations have yet to integrate AI literacy into their core curricula as per Pedró et al., 2019 [13].
- Language and cultural barriers: Much of the available AI educational content is in English, posing challenges in non-English speaking countries as per Gökçearslan et al., 2024 [18].
- Limited policy support: Many developing nations need more comprehensive policies to promote AI education and literacy as per Dutton, 2018 [11].

Despite these challenges, there are also opportunities emerging in developing nations. For instance, mobile technology penetration in many of these countries offers a potential platform for delivering AI education as per GSMA, 2020 [19]. Additionally, international collaborations and open educational resources are providing new avenues for AI literacy initiatives in resource-constrained environments as per Bonami et al., 2020 [20]. Some developing nations are making strides in AI education. For example, Rwanda has partnered with Carnegie Mellon University to establish a master's program in AI, aiming to build local capacity in this field *AI and Robotics*, n.d., [33]. Similarly, India's National Education Policy 2020 emphasizes the integration of AI education across all levels of schooling as per the Ministry of Education, Government of India, 2020 [35].

In conclusion, while the current state of AI literacy in developing nations faces significant challenges, there are also promising initiatives and opportunities emerging. Addressing these challenges will require concerted efforts from policymakers, educators, and international partners to develop tailored, culturally relevant AI literacy programs that can bridge the global AI literacy divide.

2.4 Educational Frameworks and Curriculum Development

The development of educational frameworks for AI literacy is an evolving field, with various approaches emerging globally. In developed nations, frameworks often emphasize a multidisciplinary approach. For instance, the AI4K12 Initiative in the United States proposes five big ideas for K-12 AI education: perception, representation and reasoning, learning, natural interaction, and societal impact as per Touretzky et al., 2019 [16]. This framework aims to provide a comprehensive understanding of AI, from technical concepts to ethical considerations. In Europe, the AI4EU project has developed a curriculum framework that focuses on four main areas: AI and Machine Learning foundations, AI ethics and governance, AI systems development, and AI applications as per Foffano et al., 2022 [21]. This approach seeks to balance technical skills with critical thinking about AI's societal implications. However, in developing nations, AI education frameworks are often less comprehensive and face significant implementation challenges. A study by Pedró et al., 2019 [13], found that many developing countries struggle to integrate AI concepts into their curricula due to resource constraints and a lack of tailored frameworks that consider local contexts. Current approaches to integrating AI concepts into curricula vary widely. Some countries have opted for standalone AI courses, while others integrate AI topics into existing subjects like computer science or mathematics. For example, China has introduced AI courses in primary and secondary schools, focusing on practical applications and programming skills as per Su & Yang, 2022 [22]. The effectiveness of these approaches is still being evaluated, but early studies suggest that integrated, project-based learning approaches may be more effective in developing AI literacy as per Holmes et al., 2019 [2]. However, there are significant gaps in existing frameworks, particularly for developing nations. These include a lack of focus on local context and applications, insufficient emphasis on ethical considerations specific to developing countries, and limited guidance on implementation in resource-constrained environments.

2.5 Role of Educators and Teacher Training

Educators play a crucial role in fostering AI literacy, but many face significant challenges, especially in developing nations. A survey by UNESCO, 2021 [34], found that the majority of teachers in developing countries feel unprepared to teach AI concepts. This lack of preparedness stems from several factors, including insufficient training opportunities, limited access to resources, and rapid technological changes. The importance of teacher training in AI literacy cannot be overstated. As per Chai et al., 2020 [23], argue that effective AI education requires teachers who not only understand AI concepts but can also contextualize them for their students and guide discussions on ethical implications. However, developing such expertise among teachers in resource-constrained environments presents significant challenges. In many developing nations, teachers face additional hurdles such as large class sizes, limited technology access, and competing educational priorities. A study by as per Quah, 2019 [24], in Southeast Asian countries found that while teachers recognized the importance of AI literacy, they struggled with practical implementation due to a lack

of training, insufficient infrastructure, and absence of localized teaching materials. To address these challenges, some initiatives focus on innovative teacher training approaches. For instance, the AI-Teacher project in India uses a cascading model where trained teachers become mentors for their peers, helping to scale up AI literacy education despite resource constraints as per Schiff, 2021 [12] and Chan & Zhou, 2023 [25].

3 IMPLEMENTATION STRATEGIES

The following are the proposed strategies for the developing nations.

- Various strategies have been proposed and implemented to enhance AI literacy globally. Policy initiatives play a crucial role in setting the stage for AI education. Countries like Finland have incorporated AI literacy into their national education strategies, emphasizing lifelong learning and workforce development as per Finland AI Strategy Report, n.d. [26].
- Public-private partnerships are essential for bridging the AI opportunity gap. They enhance workforce productivity through structured, role-based upskilling, create reskilling pathways using industry micro-credentials, provide high-quality education to mitigate misinformation and bias, and offer personalized learning experiences to diverse populations, breaking language barriers and making education more accessible globally as per Maggioncalda, 2024 [27].
- Online platforms and Massive Open Online Courses (MOOCs) have shown promise in democratizing access to AI education. Platforms like Coursera and edX offer AI courses from top universities, potentially reaching learners in developing nations. However, challenges remain in terms of internet access, language barriers, and adapting content to local contexts as per Bulathwela et al., 2021 [28] and Bonami et al., 2020 [20].
- The effectiveness of these strategies varies. A case study of Rwanda's partnership with Carnegie Mellon University to establish a master's program in AI demonstrates how international collaborations can build local capacity as per McSharry, 2023 [29]. However, the scalability and sustainability of such initiatives remain challenges.
- In India, the introduction of AI modules in secondary schools has shown promising results in increasing AI awareness, but faces challenges in terms of teacher preparedness and infrastructure as per Goswami & Sharma, 2024 [30].

Lessons learned from these implementations emphasize the importance of:

- Contextualizing AI education to local needs and challenges
- Focusing on teacher training and support
- Leveraging partnerships to overcome resource constraints
- Balancing technical skills with ethical and societal considerations

In conclusion, while progress has been made in developing AI literacy frameworks and implementation strategies, significant work remains to adapt these approaches to the unique contexts of developing nations. Future efforts should focus on creating flexible, culturally relevant frameworks that can be implemented with

limited resources while still providing comprehensive AI literacy education.

4 IMPACT ASSESSMENT AND MEASUREMENT

Assessing the impact of AI literacy programs is crucial for understanding their effectiveness and guiding future improvements. Existing studies employ various methods and metrics to evaluate these programs. As per Dutta & Lanvin, 2019 [31], propose a framework that measures AI literacy across four dimensions: knowledge, skills, attitudes, and behaviors. This multifaceted approach provides a comprehensive view of an individual's AI literacy. Quantitative metrics often include pre- and post-program tests to measure knowledge gains, as well as surveys to assess changes in attitudes towards AI. For instance, As per Long & Magerko, 2020 [5] used a combination of multiple-choice questions and open-ended problem-solving tasks to evaluate AI literacy levels among students. Qualitative methods, such as interviews and focus groups, are also valuable in capturing nuanced understandings and experiences. As per Kong et al., 2024 [32] employed semi-structured interviews to explore how participants applied AI concepts in real-world scenarios after completing a literacy program. Long-term impact assessment is particularly important in understanding the broader benefits of AI literacy. As per Holmes et al., 2019 [2], argue that the true value of AI literacy extends beyond immediate knowledge gains to include improved decision-making, enhanced problem-solving skills, and increased adaptability in an AI-driven world. However, conducting long-term studies in developing nations often poses challenges due to resource constraints and participant attrition.

5 NEED FOR COMPREHENSIVE FRAMEWORK

The unique challenges faced by developing nations in promoting AI literacy necessitate a tailored framework. Existing frameworks often assume a level of technological infrastructure and educational resources that may not be available in many developing countries as per Nye, 2014 [9]. Additionally, these frameworks may not adequately address the specific cultural, economic, and social contexts of developing nations.

A comprehensive framework for developing nations should consider:

1. Resource constraints and infrastructure limitations
2. Cultural and linguistic diversity
3. Local economic priorities and workforce needs
4. Existing educational systems and curricula

This paper aims to address this gap by proposing a comprehensive AI literacy framework specifically designed for developing nations. The framework will focus on essential competencies and skills, curriculum integration strategies, and the pivotal role of educator training. Furthermore, the paper will outline strategic implementation plans, taking into account the unique challenges and opportunities present in developing nations.

6 CONCLUSION & FUTURE SCOPE

This review has highlighted the growing importance of AI literacy in the global context and the significant disparities that exist between developed and developing nations. Key findings include the

multifaceted nature of AI literacy, encompassing technical knowledge, ethical considerations, and practical skills. Challenges faced by developing nations in implementing AI literacy programs include limited infrastructure, a lack of trained educators, and curriculum gaps. However, innovative strategies such as public-private partnerships and online platforms show potential in addressing these challenges. It is crucial to contextualize AI education to local needs and cultural contexts, and there is a need for comprehensive impact assessment methods to guide program improvements and demonstrate long-term benefits. To build upon this work, future research could focus on developing and validating culturally appropriate assessment tools for measuring AI literacy in diverse contexts. Conducting longitudinal studies can help understand the long-term impacts of AI literacy programs in developing nations. Additionally, exploring innovative, low-cost methods for delivering AI education in resource-constrained environments is vital. Investigating the role of AI literacy in promoting economic development and social equity in developing nations, as well as examining the intersection of AI literacy with other critical skills such as digital literacy and data science competencies, can further contribute to bridging the global AI literacy divide and ensuring that the benefits of AI technologies are accessible to all.

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